Structural Timber Framing Construction Plan

12' × 18' Rectangular Cedar Pergola Package

Project Location: Saanich, BC, Canada (BC Coastal Jurisdiction)

Design Standard: BC Building Code 2024 / Part 9 Timber Framing

Ground Snow Load: 35.0 PSF | **Wind Exposure:** Category B

Structure Type: 6-Post Rectangular Post-and-Beam Pergola with Decorative Scallop Overhangs

Material Species: Heavy Timber Yellow Cedar / Western Red Cedar

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1. Executive Summary & Design Rationale

This structural package provides complete fabrication specifications, load-path engineering, foundation schedules, and orthographic shop blueprints for a **12' × 18' rectangular heavy-timber pergola**. The layout features a symmetric 6-post grid (9' span along the 18' length and 12' clear transverse span) designed to accommodate zoned outdoor dining and lounge relaxation.

Core Dimensions & Specifications

- **Footprint:** 18'-0" (Length) × 12'-0" (Width)
- **Clear Height:** 8'-8" (9'-6" Top of Beams, 10'-0" Top of Rafters)
- **Posts (P1–P6):** 6×6 Nominal (5.5" × 5.5" actual) Yellow Cedar, 9'-6" cut length
- **Perimeter Beams (B1–B6):** 4×8 Nominal (3.5" × 7.25" actual) Yellow Cedar
- **Primary Cross-Rafters (R1–R6):** 4×6 Nominal (3.5" × 5.5" actual) with classic decorative scallop profiles

- **Lateral Knee Braces (K1A–K6B):** 4×4 Nominal (3.5" × 3.5" actual) paired 45° knee braces at all post-to-beam junctions
- **Footings (FT1–FT6):** 10" diameter Sonotube concrete piers (minimum 36" depth below frost line) with Simpson Strong-Tie ABA66Z standoff bases
- **Sun Sail Reinforcement:** Engineered tension tie points integrated into beam-post junctions

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2. Photorealistic Architectural Visualization

Below is the approved architectural rendering illustrating the completed 12' × 18' structure in a landscaped West Coast garden setting:

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3. Architectural Drawings (Presentation Layer)

Sheet A01 — Architectural Plan View

P1

P2

P3

P4

P5

P6

B1

B2

B3

B4

B5

B6

K1A

K1B

K2A

K2B

K3A

K3B

K4A

K4B

K5A

K5B

K6A

K6B

R1

R2

R3

R4

R5

R6

MAX SPAN: 18.0 FT

DRAWING PLAN VIEW

STRUCTURE: PERGOLA

JURISDICTION: BC_COASTAL

SOURCE HASH: b2c3d4e5

DATE: 2026-05-24 | SCALE: AUTO

Sheet A02 — Architectural Elevation View

P1

P2

P3

P4

P5

P6

B1

B2

B3

B4

B5

B6

K1A

K1B

K2A

K2B

K3A

K3B

K4A

K4B

K5A

K5B

K6A

K6B

R1

R2

R3

R4

R5

R6

POST: 9.5 FT

DRAWING ELEVATION VIEW

STRUCTURE: PERGOLA

JURISDICTION: BC_COASTAL

SOURCE HASH: b2c3d4e5

DATE: 2026-05-24 | SCALE: AUTO

Sheet A03 — Isometric Overview

P1

P2

P3

P4

P5

P6

B1

B2

B3

B4

B5

B6

K1A

K1B

K2A

K2B

K3A

K3B

K4A

K4B

K5A

K5B

K6A

K6B

R1

R2

R3

R4

R5

R6

OVERALL SPAN: 18.0 FT

POST HT: 9.5 FT

DRAWING ISOMETRIC VIEW

STRUCTURE: PERGOLA

JURISDICTION: BC_COASTAL

SOURCE HASH: b2c3d4e5

DATE: 2026-05-24 | SCALE: AUTO

Sheet A04 — Perspective Framing View

P1

P2

P3

P4

P5

P6

B1

B2

B3

B4

B5

B6

K1A

K1B

K2A

K2B

K3A

K3B

K4A

K4B

K5A

K5B

K6A

K6B

R1

R2

R3

R4

R5

R6

OVERALL SPAN: 18.0 FT

POST HT: 9.5 FT

DRAWING PERSPECTIVE VIEW

STRUCTURE: PERGOLA

JURISDICTION: BC_COASTAL

SOURCE HASH: b2c3d4e5

DATE: 2026-05-24 | SCALE: AUTO

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4. Orthographic Shop Blueprints (Fabrication Layer)

Sheet S01 — Engineering Foundation & Framing Plan

P1

P2

P3

P4

P5

P6

B1

B2

B3

B4

B5

B6

K1A

K1B

K2A

K2B

K3A

K3B

K4A

K4B

K5A

K5B

K6A

K6B

R1

R2

R3

R4

R5

R6

MAX SPAN: 18.0 FT

BLUEPRINT PLAN

STRUCTURE: PERGOLA

JURISDICTION: BC_COASTAL

SOURCE HASH: b2c3d4e5

DATE: 2026-05-24 | SCALE: AUTO

Sheet S02 — Structural Framing Elevation & Joint Miters

P1

P2

P3

P4

P5

P6

B1

B2

B3

B4

B5

B6

K1A

K1B

K2A

K2B

K3A

K3B

K4A

K4B

K5A

K5B

K6A

K6B

R1

R2

R3

R4

R5

R6

POST: 9.5 FT

BLUEPRINT ELEVATION

STRUCTURE: PERGOLA

JURISDICTION: BC_COASTAL

SOURCE HASH: b2c3d4e5

DATE: 2026-05-24 | SCALE: AUTO

Sheet S03 — Structural Isometric Assembly

P1

P2

P3

P4

P5

P6

B1

B2

B3

B4

B5

B6

K1A

K1B

K2A

K2B

K3A

K3B

K4A

K4B

K5A

K5B

K6A

K6B

R1

R2

R3

R4

R5

R6

OVERALL SPAN: 18.0 FT

POST HT: 9.5 FT

BLUEPRINT ISOMETRIC

STRUCTURE: PERGOLA

JURISDICTION: BC_COASTAL

SOURCE HASH: b2c3d4e5

DATE: 2026-05-24 | SCALE: AUTO

Sheet S04 — Component Details & Member Isolation

POST P1 DETAIL

6x6 Western Red Cedar / Douglas Fir (9.5 FT)

POST HT: 9.5 FT

5.5 IN NOM

BEAM B1 DETAIL

4x8 Western Red Cedar / Douglas Fir (9.00 FT, Miter 30.00°)

BEAM MITER: 30.00°

SPAN 9.00 FT

RAFTER R1 DETAIL

4x6 Western Red Cedar / Douglas Fir (10.00 FT, Pitch 0:12)

PITCH 0:12 | MITER 28.71°

LEN 10.00 FT

KNEE BRACE K1A DETAIL

4x4 Western Red Cedar / Douglas Fir (45.00° Miter)

45.00° MITER

RUN 1.5 FT

BLUEPRINT COMPONENT ISOLATION

STRUCTURE: PERGOLA

JURISDICTION: BC_COASTAL

SOURCE HASH: b2c3d4e5

DATE: 2026-05-24 | SCALE: AUTO

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5. Bill of Materials & Cut Schedules

Refer to the accompanying standalone builder documentation for complete breakdown:

- **Cut List:** `outputs/fabrication/cut-list.json` & `outputs/shop-blueprint/SB01-cut-list.json`
- **Lumber Purchase Order:** `outputs/lumber-purchase-list.md`
- **Budget Estimate:** `outputs/budget-estimate.md`
- **Assembly Guide:** `outputs/assembly-guide.md`